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**FRY FOUNDATION
DECENTRALIZED VPN
SYSTEM (DVPN)**

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1. INTRODUCTION

1.1 Purpose

The purpose of this document is to outline the development requirements for a decentralized Virtual Private Network (DVPN) that leverages bandwidth miners as network endpoints. This innovative DVPN will use cryptocurrency, specifically FRY, to handle subscription payments and integrate a Proof of Connectivity (PoC) mechanism to verify active contributions to the network. The project aims to enhance both the security and efficiency of traditional VPNs, creating a secure, decentralized alternative for users.

Additionally, the DVPN will improve upon an existing system by adding advanced authentication features, notably through the use of Algorand wallet addresses for user identification and payment. This document will cover the technical scope, proposed enhancements, and required functionalities to ensure the VPN service is not only reliable and secure but also provides a seamless experience for all users.

This document outlines the development of a decentralized VPN (DVPN) powered by user-contributed bandwidth, with cryptocurrency (FRY) handling payments and Proof of Connectivity (PoC) ensuring network reliability. The goal is to create a secure, efficient VPN that verifies users through Algorand wallets.

1.2 Vision

The vision of this project is to establish a secure, decentralized VPN network driven by user-contributed bandwidth. By participating in the network, users are rewarded with FRY cryptocurrency, incentivizing a collaborative approach to internet security and privacy.

This decentralized model empowers users worldwide, making reliable, private internet access more accessible and transparent. Leveraging blockchain technology, the project aims to create a VPN network that protects user privacy without relying on central entities, promoting a truly user-controlled internet experience.

2. OVERALL DESCRIPTION

2.1 System Overview

This decentralized VPN system allows users to contribute their internet bandwidth to the network, earning FRY cryptocurrency as a reward. The system relies on blockchain technology to ensure secure and private access to the internet, managed directly by users without central control.

2.1.1 Product Functions

- **User Authentication:** Allows secure login through Algorand wallet addresses, streamlining user access with a focus on blockchain-based validation for secure and efficient authentication.
- **Subscription Management:** Offers a cryptocurrency-based subscription system where users pay in Frye for VPN services. Options include monthly and yearly subscriptions, and the system tracks and manages payments.
- **Peer-to-Peer Connection:** Enables direct connections between users and bandwidth miners on the network, which optimizes network resource distribution by routing data through available miners.
- **Proof of Connectivity:** Verifies network contributions by miners via automated FRY transactions, ensuring users' bandwidth contributions are continuously validated.
- **Error Handling and QA:** Implements quality assurance processes to identify and resolve issues like connection errors, UI bugs, and potential security risks, promoting system stability and reliability.

2.1.2 System Features

- **P2P VPN Connectivity:** Provides a secure, peer-to-peer VPN connection, allowing users to connect through available network endpoints instead of centralized servers.
- **Bandwidth Mining Integration:** Integrates a system for bandwidth mining, where users share network resources and earn rewards based on contributions.
- **Cryptocurrency-Based Subscription System:** Uses the Frye cryptocurrency for subscription payments, making the VPN service accessible with crypto transactions.
- **Algorand Wallet Authentication:** Authenticates users via their Algorand wallet, reinforcing secure access with blockchain-based validation.
- **Global Endpoint Network:** Expands the VPN network globally through miners who share their bandwidth, offering a more distributed and resilient VPN structure.
- **Proof of Connectivity Mechanism:** Verifies that miners actively contribute bandwidth by validating their connections with automated checks and transactions.
- **Automated Reward Distribution System:** Ensures users who share their bandwidth receive rewards, automatically distributing Frye cryptocurrency based on verified connectivity.

2.2 User Classes and Characteristics

- **End Users:** Individuals who subscribe to the VPN service, accessing secure connections and earning rewards through platform interactions.
- **Bandwidth Miners:** Users contributing bandwidth to the network, playing a critical role in P2P connectivity and rewarded for active participation.
- **Administrators:** Responsible for managing user accounts, handling subscriptions, overseeing network health, ensuring reward distribution, and maintaining overall security.

2.3 Assumptions and Dependencies

- **Dependencies:** The application relies on Algorand blockchain integration for authentication and payment, the Frye cryptocurrency for transactions, and a stable internet connection for P2P connectivity.
- **Assumptions:** Users are expected to have Algorand wallets for authentication, reliable internet access for VPN and mining, and compatible devices to support bandwidth contribution and P2P network participation.

3. SPECIFIC REQUIREMENT

3.1 Functional Requirements

3.1.1 VPN Connectivity

- Establishes secure connections between clients and bandwidth miners.
- Supports endpoint selection based on geographical location.
- Ensures stable connections, reducing disconnections.
- Provides real-time connection status and metrics for users.

3.1.2 Authentication System

- Supports user authentication through Algorand wallet addresses.
- Implements wallet ownership verification for added security.
- Manages secure sessions to maintain user data integrity.
- Includes both client and server authentication mechanisms.

3.1.3 Payment System

- Implements a subscription-based payment system using FRY cryptocurrency.
- Supports both monthly and yearly subscription models for flexibility.
- Verifies payment status before granting VPN access.
- Maintains payment history and subscription statuses for users.
- Manages an automatic renewal process for ongoing subscriptions.

3.1.4 Bandwidth Mining Integration

- Integrates with bandwidth mining devices as part of the network.
- Registers new miners as network endpoints for connectivity.
- Monitors miner availability and performance to ensure service quality.
- Distributes client connections among available miners.

3.1.5 Proof of Connectivity (PoC) System

- The system implements an automated Proof of Connectivity (PoC) mechanism within the server application.
- Sends 0 FRY transactions hourly from the user's Algorand address to verify active participation in the dVPN network.
- Maintains a record of all PoC transactions for accurate tracking of connectivity and participation.
- Integrates with the existing reward distribution script to calculate rewards based on verified PoC transactions.
- Manages transaction failures by incorporating error-handling and retry mechanisms, improving reliability.
- Validates each transaction to confirm successful submission.
- Provides monitoring and reporting capabilities to track PoC status and network health.

3.1.6 Reward Distribution

- Tracks the number of 0 FRY transactions per device over a 24-hour period.
- Integrates with the reward distribution script for seamless payout.
- Maintains accurate records of rewards and distributions.
- Manages distribution failures with retries to ensure consistency.
- Provides reporting on the status of reward distributions.

3.2 Non-Functional Requirements

3.2.1 Performance Requirements

The VPN should maintain high speed and low latency for users, with minimal interruptions. The system must also handle multiple connections efficiently.

3.2.2 Security Requirements

Strong encryption and blockchain security measures are essential to protect user data, payment details and maintain the integrity of rewards.

3.2.3 Usability Requirements

The platform should have an intuitive and user-friendly interface, making it simple for users to navigate, connect to the VPN, and monitor rewards.

4. PROPOSED SOLUTION

1. VPN Connectivity

1.1 Secure VPN Connections

- We will establish secure VPN connections between clients and bandwidth miners.
- Encryption will protect data privacy and ensure secure data transmission.
- Regular security assessments will be conducted to identify and address vulnerabilities.

1.2 Geographical Endpoint Selection

- Users will be able to choose VPN endpoints based on their location preferences.
- The interface will allow users to easily browse bandwidth miners by location.
- An intelligent algorithm will select endpoints with the best performance to ensure a smooth experience.

1.3 Maintaining Stable Connections

- Tools will monitor connections to prevent disconnections and enhance stability.
- Adaptive quality management will adjust to network conditions to maintain stability.
- Extensive testing will validate stable connections across various network scenarios.

1.4 Real-Time Connection Status

- Users will have a dashboard to view real-time connection status and performance metrics.
- Key metrics like speed, latency, and uptime will be shown clearly for transparency.
- Alerts will notify users of any connection issues or required actions.

2. Authentication System

2.1 Algorand Wallet-Based User Authentication

- We will enable user authentication through Algorand wallet addresses.
- Users can sign up with their existing Algorand wallets, making registration quick and simple.
- Clear instructions will guide users to connect their wallets seamlessly.

2.2 Wallet Ownership Verification

- A wallet verification step will confirm users control their wallets.
- Users will sign a unique message to prove ownership, with immediate feedback on success or failure.

2.3 Secure Session Management

- Secure session management will protect user interactions after login.
- Session tokens will be used and will include expiration and refresh features for added security.

2.4 Client and Server Authentication

- Both client and server authentication will secure user connections.
- Client identity verification will prevent unauthorized access.
- Multi-factor authentication may be added as an extra security option.

3. Payment System

3.1 FRY Cryptocurrency Subscription Integration

- Users will be able to subscribe monthly or yearly using FRY cryptocurrency.
- A secure payment gateway will handle transactions, validating each one on the FRY blockchain.
- Automatic checks will verify payment status before granting VPN access.

3.2 Flexible Subscription Models

- Monthly and yearly subscription plans will be offered for user flexibility.
- Renewal reminders will prompt users as subscriptions approach expiration.
- Real-time subscription status updates will ensure users always have access in line with their payments.

3.3 Payment Status Verification

- Real-time checks will confirm payments before granting VPN access.
- A system will track payment confirmations on the FRY blockchain to update user access.
- Error handling will provide smooth experiences even if payment processing issues arise.

3.4 Payment History and Subscription Status

- Users can access their payment history, including transaction details, directly within the app.
- A user interface will allow users to manage their subscriptions and view their payment status securely.

3.5 Automatic Renewal Process

- Users can opt into automatic renewals, which will process payments before subscriptions expire.
- Renewal reminders will notify users in advance, with options to update payment information.

3.6 Integration with Authentication

- Payment status will link directly to user accounts, ensuring secure handling of subscription data.
- Users with unpaid subscriptions will be restricted from VPN access until payments are complete.

4. Bandwidth Mining Integration

4.1 Device Integration

- The system will support seamless integration with bandwidth mining devices for a decentralized network.
- Communication protocols will be established for reliable interaction between miners and the DVPN app.
- Compatibility updates will be provided to support various mining devices.

4.2 Miner Registration

- A simplified registration process will make it easy for new miners to join the network.
- Validation mechanisms will verify and authenticate new miners before they become active endpoints.

4.3 Monitoring Miner Performance

- Monitoring tools will track each miner's availability, uptime, and bandwidth quality.
- Real-time analytics will allow administrators to monitor performance and receive alerts for issues.

4.4 Load Balancing

- An algorithm will distribute client connections evenly across available miners.
- Miner performance and client loads will be continuously analyzed for optimal resource use.

5. Proof of Connectivity (PoC)

5.1 Automated PoC Transactions

- An automated system will send hourly 0 FRY transactions from users' Algorand addresses.
- The system will log all PoC transactions for tracking and troubleshooting.

5.2 Transaction Management

- A queue system will manage and schedule transactions, with retries for failed transactions.
- Duplicate transactions will be prevented by tracking recent activity.

5.3 Reward Distribution Integration

- Transaction data will integrate with the reward distribution script.
- Daily transaction summaries will track rewards per device, with monitoring for any failed distributions.

5.4 Performance Optimization

- The PoC system will use minimal resources while ensuring reliability.
- Batch processing and efficient data structures will improve transaction monitoring and scalability.

5.5 Monitoring and Reporting

- A dashboard will show real-time PoC activity, including success rates and reward distribution.
- Automated alerts and regular reports will summarize performance and flag any issues.

5.6 Security Measures

- Secure key management will be in place for signing transactions.
- Access controls and detailed logging will safeguard PoC activities.

5. TECHNOLOGY STACK

5.1 Design

- **Figma:** Used for designing the user interface (UI) and user experience (UX). Figma supports collaborative, real-time design, allowing teams to work together efficiently on wireframes, prototypes, and high-fidelity design elements.

5.2 Frontend

- **WalletConnect:** Enables seamless, secure connections between decentralized wallets and the application, providing a safe and convenient user interface for wallet-based interactions.

5.3 Backend

- **libp2p:** Powers peer-to-peer networking, allowing nodes to connect and communicate securely and efficiently within the decentralized network.
- **WireGuard:** A VPN protocol integrated for its high-speed encryption capabilities, providing secure data transmission across backend services.

5.4 Desktop Application Development

- **C#:** Primary language for developing the desktop application, providing a rich environment for building user interfaces and handling client-side logic on desktop.

5.5 Storage

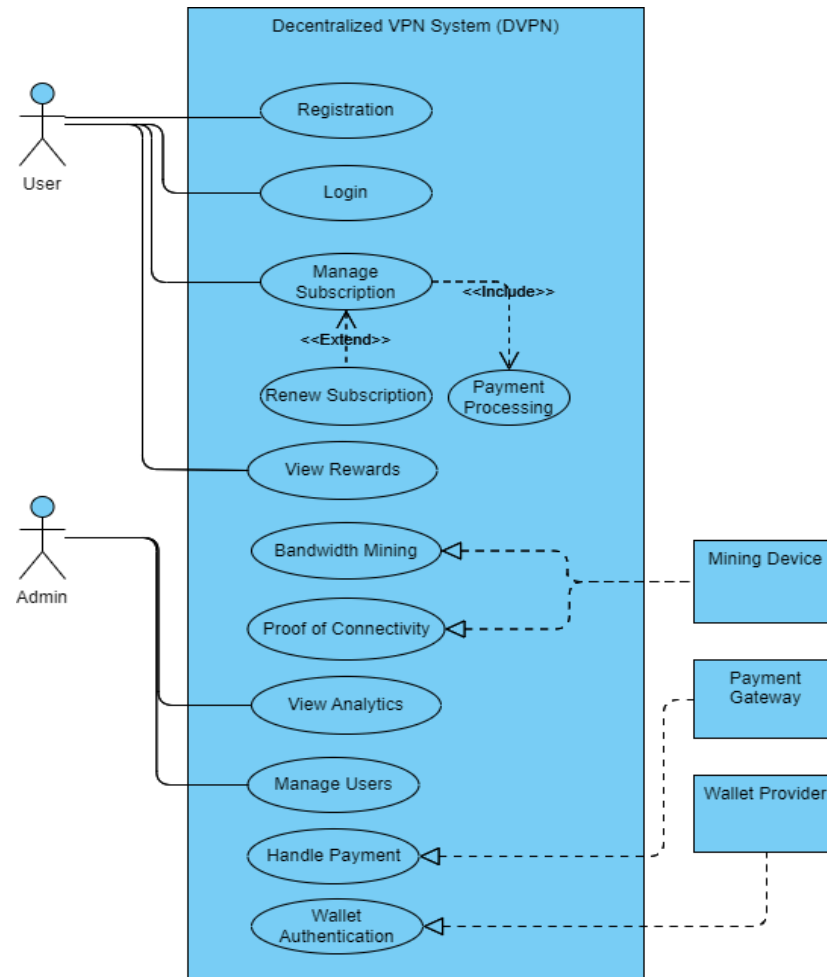
- **IPFS (InterPlanetary File System):** A decentralized file storage system used for distributing and retrieving data efficiently, ensuring data redundancy and availability across the network.

5.6 Blockchain

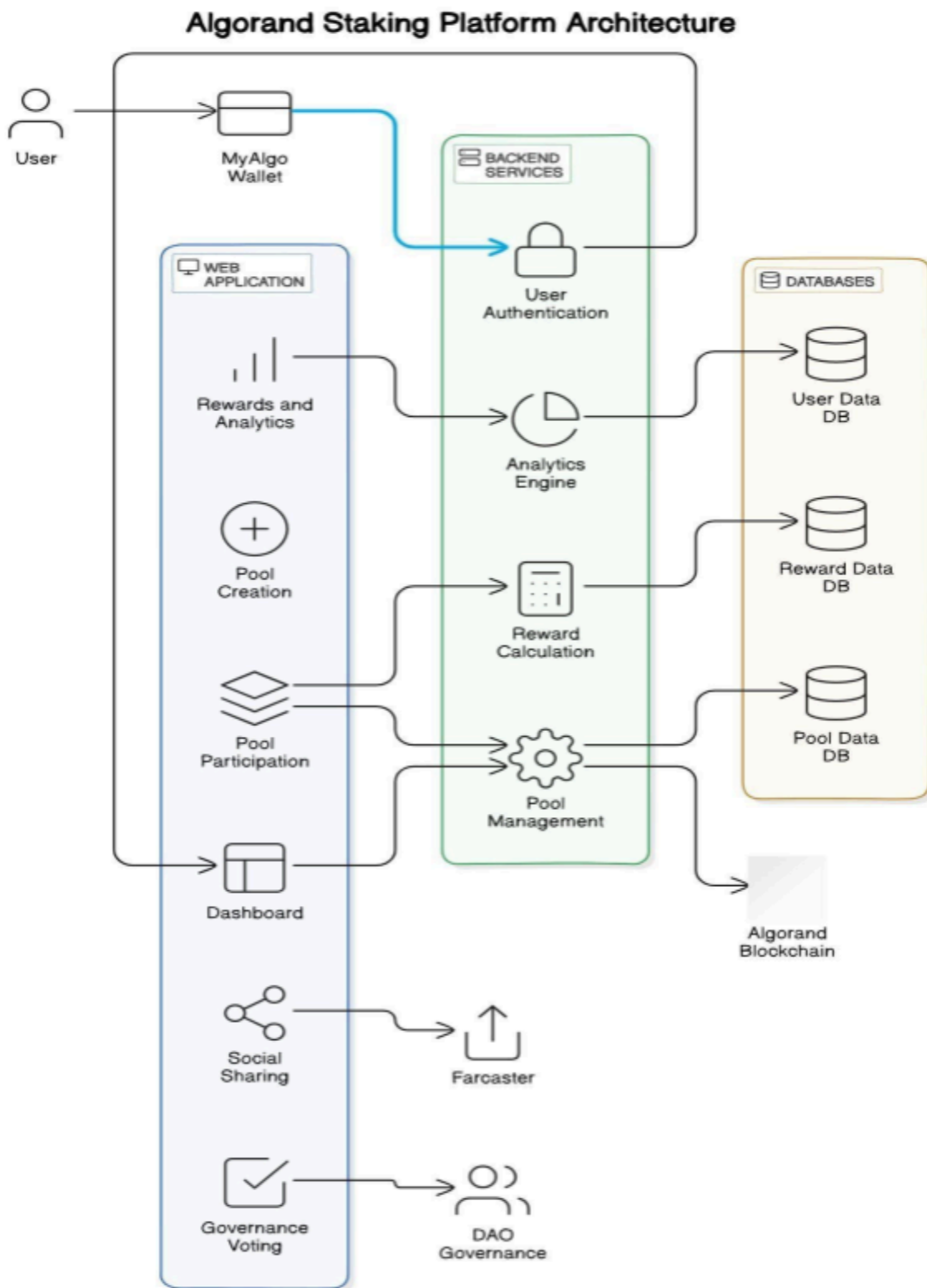
- ***PyTeal***: A Python library for Algorand smart contracts, allowing secure contract development for platform functionalities, particularly for transaction and reward distribution.

6. DESIGN SPECIFICATIONS

6.1 Use Case Diagram



6.2 System Diagram



7. DEVELOPMENT PLAN

7.1 Implementation Phases

7.1.1 Phase 1: Initial Setup and Integration

Set up the project's core infrastructure, including database and server configuration, VPN connectivity, authentication setup, and payment integration.

7.1.2 Phase 2: Functional Component Development

Develop core features such as VPN connection, payment processing, bandwidth mining, and reward distribution systems, ensuring each function operates as intended.

7.1.3 Phase 3: Security Enhancements

Strengthen security measures across all components, including data encryption, secure session handling, and wallet authentication to protect user data and system integrity.

7.1.4 Phase 4: Performance Optimization

Refine the application for improved performance by optimizing server response times, ensuring connection stability, and monitoring the system for bottlenecks.

7.2 Project Timeline and Milestones

Establish a timeline with key milestones for each development phase, ensuring the project stays on track. Track progress through specific dates and deliverables for each phase to manage project completion effectively.

7.3 Resource Allocation

Assign resources to each development phase, ensuring adequate support in terms of developer expertise, security, and design resources to meet project requirements efficiently and within the planned timeline.

8. SUBSCRIPTION PLAN

Monthly and Yearly Subscription Options	● Monthly Subscription: Users can subscribe on a month-to-month basis, paying with FRY cryptocurrency. This plan is ideal for those who prefer short-term access. ● Yearly Subscription: Users can opt for a discounted annual subscription, offering cost savings for long-term access.
FRY Cryptocurrency Integration	● Payment Gateway: Transactions are processed through a secure FRY cryptocurrency payment gateway, ensuring seamless and secure payments. ● Payment Validation: The system verifies each transaction on the FRY blockchain to confirm payment status before granting access to VPN services. ● Automated Payment Handling: The application automatically updates the subscription status upon successful FRY transaction verification.
Automatic Renewal Process	● Opt-In Feature: Users can opt-in for automatic renewals when signing up, allowing seamless continuation of their subscription without manual renewals. ● Renewal Reminders: Notifications remind users of upcoming renewals, with the option to update payment details if needed. ● Automated Payment Requests: For users enrolled in auto-renewal, the system automatically initiates a

	renewal transaction through the FRY payment gateway before expiration.
Real-Time Subscription Status Updates	● Payment Status Verification: The system checks payment status in real-time before granting VPN access, ensuring only valid subscribers have access. ● Automated Access Control: Access is restricted if a payment is pending, with prompts for the user to complete the subscription process.
User-Friendly Subscription Management	● Subscription Dashboard: Users can view their subscription history, including payment dates, amounts, transaction status, and subscription duration. ● Manage Subscription Status: Users can renew, update, or cancel subscriptions directly within the DVPN application for full control over their plan.
Seamless Authentication Integration	● Link with Algorand Wallets: The payment system integrates with Algorand-based user authentication, ensuring that each FRY transaction is securely linked to the user's account. ● Account-Based Access Control: The subscription status is directly tied to the user's Algorand wallet address, simplifying secure access and maintaining data privacy.

9. TESTING AND QUALITY ASSURANCE

9.1 Testing Strategy

9.1.1 Functional Testing

Functional testing will verify each feature's performance against the specified requirements. Tests will focus on verifying the correct functionality of VPN connectivity, authentication, payment processing, and reward distribution to ensure the system meets all intended use cases.

9.1.2 Security Testing

Security testing will focus on identifying vulnerabilities within the application's components, including VPN connection protocols, wallet authentication, and payment handling. This approach will include simulated threat scenarios to verify the strength of data protection, encryption, and secure session management.

9.1.3 Usability Testing

Usability testing will be implemented to assess the user experience by evaluating the intuitiveness of navigation, clarity of the interface, and ease of access to features. User feedback will be analyzed to ensure the system meets usability standards, facilitating adjustments to enhance user satisfaction.

9.2 Test Cases and Scenarios

Detailed test cases and scenarios will cover each functional and non-functional requirement, ensuring comprehensive testing across all components. This includes cases for typical user actions, security challenges, and error scenarios, establishing a structured framework to verify expected outcomes.

9.3 Performance Testing

Performance testing will evaluate the system's response times, stability, and scalability. Tests will be executed under various load conditions to measure the efficiency of bandwidth mining, VPN connectivity, and payment processing, allowing for real-time optimizations to improve system speed and reliability.

9.4 Error Handling and Validation

The system's error handling will be tested by simulating erroneous inputs and unexpected actions. Validation checks will ensure that any invalid data is appropriately flagged, while graceful failure protocols will be confirmed to maintain functionality during adverse conditions.

10. SECURITY CONSIDERATIONS

10.1 Data Encryption and Storage

Data encryption will be applied to protect sensitive information during transmission and storage. Encryption protocols will ensure that user data, payment information, and system records remain secure and inaccessible to unauthorized parties.

10.2 Secure Key Management

Secure key management practices will be implemented to handle encryption keys effectively, ensuring that keys are stored securely and access is restricted. This will prevent unauthorized access to encrypted data, maintaining system integrity.

10.3 Access Control Mechanisms

Access control mechanisms will enforce strict user authentication and authorization. This will ensure that only verified users and administrators can access specific functionalities, adding an extra layer of security to the application.

10.4 Transaction Validation

Transaction validation processes will confirm the authenticity and integrity of each transaction, using mechanisms like proof-of-connectivity (PoC) and secure payment verification. This will ensure that all transactions are legitimate and protect against fraudulent activities.

11. MONITORING AND REPORTING

11.1 Real-Time Monitoring Tools

Real-time monitoring tools will be used to continuously track system performance, ensuring that any issues are identified and addressed immediately. These tools provide insights into active connections, resource usage, and overall system health.

11.2 Reporting Metrics and Analytics

Metrics and analytics will be collected to track key system indicators, such as user activity, transaction volumes, and performance statistics. This data will help in understanding system usage patterns and identifying areas for improvement.

11.3 Alerts and Notifications for System Anomalies

Alerts and notifications will be set up to detect unusual activity or potential issues, such as connectivity failures or security threats. Immediate notifications allow the team to respond quickly to maintain system stability and security.

11.4 Periodic Reports and System Audits

Periodic reports and system audits will be conducted to review system health, ensure compliance with security standards, and verify that all components function as expected. Regular audits will contribute to overall system reliability and data integrity.

12. CONCLUSION

This document outlines a comprehensive plan for building a secure, efficient, and user-friendly platform. Each section details key aspects, from system requirements and design to testing, security, and monitoring strategies. By following this structured approach, we aim to ensure a smooth development process, resulting in a high-quality product that meets user needs and maintains strong performance and security standards. This roadmap supports our commitment to delivering a reliable and effective solution for all users.

13. GLOSSARY

Authentication System	A security process that verifies user identities, ensuring only authorized access to the platform.
Bandwidth Mining	The process of using a device's available bandwidth for data transfers, potentially earning rewards.
Connection Stability	Maintaining a consistent and reliable connection to prevent disruptions in service.
Data Encryption	The process of converting information into a secure format, protecting it from unauthorized access.
Endpoint Selection	Choosing the geographical location of a server or node to connect with, impacting connection speed and security.
FRY Cryptocurrency	A native digital currency used for transactions and subscriptions within the platform.
IPFS (InterPlanetary File System)	A distributed file system that allows for secure storage and sharing of files across multiple locations.
libp2p	A network framework used to enable peer-to-peer connections within the platform.
Monitoring Tools	Software that tracks system performance and detects issues in real time.
Payment System	The infrastructure that manages user payments, subscription fees, and transaction history.
Proof of Connectivity (PoC)	A system that verifies active connections to reward users based on consistent and valid connectivity.
Secure Key Management	Practices that protect cryptographic keys used for data encryption and system authentication.
System Audit	A periodic review of the platform to ensure compliance with security and operational standards.
Transaction Validation	The process of confirming and recording each transaction's legitimacy within the platform.

Usability Testing	A testing method to assess the platform's user-friendliness and ensure a smooth experience.
VPN (Virtual Private Network)	A tool for creating a secure, encrypted connection over a less secure network, often used for privacy and security.
WalletConnect	A protocol allowing users to connect their cryptocurrency wallets to decentralized applications for secure transactions.